Experiment 9: Named Entity Recognition (NER)

Aim

To implement **Named Entity Recognition (NER)** and identify named entities such as person, location, and organization in a given text.

Theory

Named Entity Recognition (NER) is a task in **Natural Language Processing (NLP)** that involves identifying and classifying important entities in text into predefined categories.

Example:

- *"Elon Musk is the CEO of Tesla."*
  → Person: Elon Musk
  → Organization: Tesla

Methods of NER

- **Rule-Based Approach:**
  Uses predefined rules and patterns (like capitalization). Simple but less flexible.
- **Supervised Learning:**
  Uses labeled datasets and algorithms like CRF or SVM. More accurate but needs training data.
- **Deep Learning:**
  Uses models like LSTM and BERT for better context understanding. Highly accurate but computationally expensive.

Conclusion

Named Entity Recognition helps in extracting useful information from text by identifying important entities. In this experiment, different approaches are studied, where rule-based methods are simple but limited, while machine learning and deep learning methods provide better accuracy. The choice of method depends on data availability and application requirements.